Bayesian Analysis (2024) **TBA**, Number TBA, pp. 1

A sample article title

First Author*, Second Author† and Third Author†

Abstract. The abstract should summarize the contents of the paper. It should be clear, descriptive, self-explanatory and not longer than 200 words. It should also be suitable for publication in abstracting services. Please avoid using math formulas as much as possible.

This is a sample input file. Comparing it with the output it generates can show you how to produce a simple document of your own.

MSC2020 subject classifications: Primary 00X00, 00X00; secondary 00X00.

Keywords: First keyword, second keyword.

1 Introduction

This sample helps you to create a properly formatted LaTeX 2_ε manuscript. Prepare your paper in the same style as used in this sample .pdf file. Try to avoid excessive use of italics and bold face. Please do not use any LaTeX 2_ε or TeX commands that affect the layout or formatting of your document (i.e., commands like `\textheight`, `\textwidth`, etc.).

2 Section headings

Here are some sub-sections:

2.1 A sub-section

Regular text.

A sub-sub-section

Regular text.

3 Text

3.1 Lists

The following is an example of an *itemized* list, two levels deep.

arXiv: 2010.00000

*Department, University or Company Name, first@somewhere.com

†Department, University or Company Name, second@somewhere.com; third@somewhere.com

© 2024 International Society for Bayesian Analysis

- This is the first item of an itemized list. Each item in the list is marked with a "tick." The document style determines what kind of tick mark is used.
- This is the second item of the list. It contains another list nested inside it.
  - This is the first item of an itemized list that is nested within the itemized list.
  - This is the second item of the inner list. LaTeX allows you to nest lists deeper than you really should.

  This is the rest of the second item of the outer list.
- This is the third item of the list.

The following is an example of an *enumerated* list of one level.

1. This is the first item of an enumerated list.
2. This is the second item of an enumerated list.

The following is an example of an *enumerated* list, two levels deep.

1. This is the first item of an enumerated list. Each item in the list is marked with a "tick." The document style determines what kind of tick mark is used.
2. This is the second item of the list. It contains another list nested inside of it.
   (a) This is the first item of an enumerated list that is nested within.
   (b) This is the second item of the inner list. LaTeX allows you to nest lists deeper than you really should.

   This is the rest of the second item of the outer list.
3. This is the third item of the list.

3.2 Punctuation

Dashes come in three sizes: a hyphen, an intra-word dash like "U-statistics" or "the time-homogeneous model"; a medium dash (also called an "en-dash") for number ranges or between two equal entities like "1–2" or "Cauchy–Schwarz inequality"; and a punctuation dash (also called an "em-dash") in place of a comma, semicolon, colon or parentheses—like this.

Generating an ellipsis . . . with the right spacing around the periods requires a special command.

4 Fonts

Please use text fonts in text mode, e.g.:

Roman

Italic

Bold

Small Caps

Sans serif

`Typewriter`

Please use mathematical fonts in mathematical mode, e.g.:

ABCabc123

$\mathit{ABCabc123}$

$\mathbf{ABCabc123}$

$\boldsymbol{ABCabc123\alpha\beta\gamma}$

$\mathcal{ABC}$

$\mathbb{ABC}$

$\mathsf{ABCabc123}$

$\mathtt{ABCabc123}$

$\mathfrak{ABCabc123}$

Note that `\mathcal, \mathbb` belongs to capital letters-only font typefaces.

5 Notes

Footnotes[1] pose no problem.[2]

[1] This is an example of a footnote.
[2] Note that footnote number is after punctuation.

6 Quotations

Text is displayed by indenting it from the left margin. There are short quotations

> This is a short quotation. It consists of a single paragraph of text. There is no paragraph indentation.

and longer ones.

> This is a longer quotation. It consists of two paragraphs of text. The beginning of each paragraph is indicated by an extra indentation.
>
> This is the second paragraph of the quotation. It is just as dull as the first paragraph.

7 Environments

7.1 Examples for *`plain`*-style environments

Axiom 1. *This is the body of Axiom 1.*

Proof. This is the body of the proof of the axiom above. □

Claim 2. *This is the body of Claim 2. Claim 2 is numbered after Axiom 1 because we used* `[axiom]` *in* `\newtheorem`*.*

Theorem 7.1. *This is the body of Theorem 7.1. Theorem 7.1 numbering is dependent on section because we used* `[section]` *after* `\newtheorem`*.*

Theorem 7.2 (Title of the theorem)**.** *This is the body of Theorem 7.2. Theorem 7.2 has additional title.*

Lemma 7.3. *This is the body of Lemma 7.3. Lemma 7.3 is numbered after Theorem 7.2 because we used* `[theorem]` *in* `\newtheorem`*.*

Proof of Lemma 7.3. This is the body of the proof of Lemma 7.3. □

7.2 Examples for *`definition`*-style environments

Definition 7.4. This is the body of Definition 7.4. Definition 7.4 is numbered after Lemma 7.3 because we used `[theorem]` in `\newtheorem`.

Example. This is the body of the example. Example is unnumbered because we used `\newtheorem*` instead of `\newtheorem`.

Fact. This is the body of the fact. Fact is unnumbered because we used `\newtheorem*` instead of `\newtheorem`.

7.3 Example for *`remark`*-style environments

Case 1. This is the body of Case 1.

Table 1: The spherical case ($I_1 = 0$, $I_2 = 0$)

Equil. points	x	y	z	C	S
L_1	−2.485252241	0.000000000	0.017100631	8.230711648	U
L_2	0.000000000	0.000000000	3.068883732	0.000000000	S
L_3	0.009869059	0.000000000	4.756386544	−0.000057922	U
L_4	0.210589855	0.000000000	−0.007021459	9.440510897	U
L_5	0.455926604	0.000000000	−0.212446624	7.586126667	U
L_6	0.667031314	0.000000000	0.529879957	3.497660052	U
L_7	2.164386674	0.000000000	−0.169308438	6.866562449	U
L_8	0.560414471	0.421735658	−0.093667445	9.241525367	U
L_9	0.560414471	−0.421735658	−0.093667445	9.241525367	U
L_{10}	1.472523232	1.393484549	−0.083801333	6.733436505	U
L_{11}	1.472523232	−1.393484549	−0.083801333	6.733436505	U

Table 2: Sample posterior estimates for each model

				Quantile		
Model	Parameter	Mean	Std. dev.	2.5%	50%	97.5%
Model 0	β_0	−12.29	2.29	−18.04	−11.99	−8.56
	β_1	0.10	0.07	−0.05	0.10	0.26
	β_2	0.01	0.09	−0.22	0.02	0.16
Model 1	β_0	−4.58	3.04	−11.00	−4.44	1.06
	β_1	0.79	0.21	0.38	0.78	1.20
	β_2	−0.28	0.10	−0.48	−0.28	−0.07
Model 2	β_0	−11.85	2.24	−17.34	−11.60	−7.85
	β_1	0.73	0.21	0.32	0.73	1.16
	β_2	−0.60	0.14	−0.88	−0.60	−0.34
	β_3	0.22	0.17	−0.10	0.22	0.55

8 Tables and figures

Cross-references to labeled tables: As you can see in Table 1 and also in Table 2.

Sample of cross-reference to figure. Figure 1 shows that it is not easy to get something on paper.

9 Equations and the like

Two equations:

$$C_s = K_M \frac{\mu/\mu_x}{1 - \mu/\mu_x} \tag{1}$$

and

$$G = \frac{P_{\text{opt}} - P_{\text{ref}}}{P_{\text{ref}}} 100(\%). \tag{2}$$

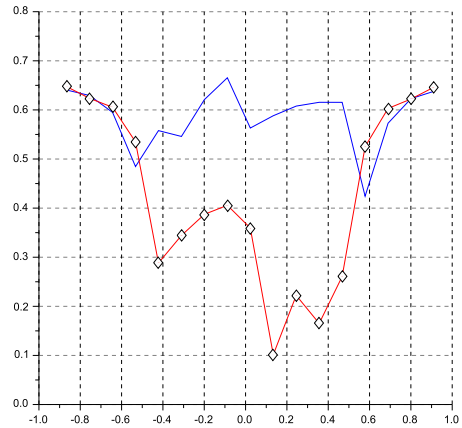

Figure 1: Pathway of the penicillin G biosynthesis.

Equation arrays:

$$\frac{dS}{dt} = -\sigma X + s_F F, \tag{3}$$
$$\frac{dX}{dt} = \mu X, \tag{4}$$
$$\frac{dP}{dt} = \pi X - k_h P, \tag{5}$$
$$\frac{dV}{dt} = F. \tag{6}$$

One long equation:

$$\begin{aligned} \mu_{\text{normal}} &= \mu_x \frac{C_s}{K_x C_x + C_s} \\ &= \mu_{\text{normal}} - Y_{x/s}\big(1 - H(C_s)\big)(m_s + \pi/Y_{p/s}) \\ &= \mu_{\text{normal}}/Y_{x/s} + H(C_s)(m_s + \pi/Y_{p/s}). \end{aligned} \tag{7}$$

The following are some other citations to check that the bibliography style file is working correctly: Akaike [1], Ben-Akiva and Lerman [2], Dyke and Patterson [3], Greene [5], Kendall and Stuart [9], Hilbe [7], Hilbe [8], Hilbe [6], Maddala [10], and Goossens, Mittelbach, and Samarin [4].

Acknowledgments

The authors would like to thank the anonymous referees, an Associate Editor and the Editor for their constructive comments that improved the quality of this paper.

Funding

The first author was supported by NSF Grant DMS-??-??????.

The second author was supported in part by NIH Grant ??????????.

Supplementary Material

Title of Supplement A. Short description of Supplement A.

Title of Supplement B. Short description of Supplement B.

References

[1] Akaike, H. (1973). "Information theory and an extension of the maximum likelihood principle." In Petrov, B. N. and Csaki, F. (eds.), *Second International Symposium on Information Theory*, 267–281. Budapest, Hungary: Akademiai Kiado. 6

[2] Ben-Akiva, M. and Lerman, S. R. (1985). *Discrete Choice Analysis: Theory and Application to Travel Demand*. Cambridge, MA: MIT Press. 6

[3] Dyke, G. V. and Patterson, H. D. (1952). "Analysis of factorial arrangements when the data are proportions." *Biometrics*, 8: 1–12. 6

[4] Goossens, M., Mittelbach, F., and Samarin, A. (1994). *The LaTeX Companion*. Reading, MA: Addison–Wesley. 6

[5] Greene, W. (2000). *Econometric Analysis*. Upper Saddle River, NJ: Prentice–Hall, 4th edition. 6

[6] Hilbe, J. (1993a). "Generalized linear models." In *Bayesian Analysis Technical Bulletin Reprints*, volume 2, 149–159. College Station, TX: Bayesian Analysis Press. 6

[7] — (1993b). "Log Negative Binomial Regression as a Generalized Linear Model." Technical Report Technical Report 26, Graduate College Committee on Statistics. 6

[8] — (1994). "Generalized linear models." *The American Statistician*, 48(3): 255–265. 6

[9] Kendall, M. and Stuart, A. (1979). *The Advanced Theory of Statistics*, volume 2. London: Charles Griffin & Company, 4th edition. 6

[10] Maddala, G. S. (1992). *Introduction to Econometrics*. New York: MacMillan, 2d edition. 6